SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 30%

QUESTIONS: 3 **Total Marks:30**

Annexure C

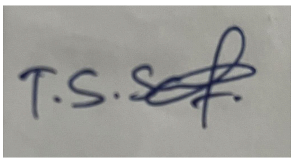
Case study

Criteria	Understanding of Case ***(2.5)***	Application of Concepts ***(2.5)***	Analysis ***(2)***	Solution Design ***(2)***	Clarity ***(1)***	***Total(10)***
Weightage	*25%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						

Code of Conduct:

I T.S.Sabarish 192371014 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANNEXURE A

Case Study:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP,FP,FN,TN

2. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP,FP,FN,TN and Calculate the Precision and Recall and F1 Score .

3. An email service provider wants to classify emails as **Spam or Not Spam** using features:

- Contains “Offer” (Yes/No)
- Contains “Free” (Yes/No)
- Email Length (Short/Long)

Sample Dataset

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:
   - P(Spam), P(Not Spam)
2. Compute conditional probabilities for each feature.
3. Using **Naïve Bayes**, classify:
   - (Offer=Yes, Free=No, Length=Short)
4. Compare results using **WEKA (Naïve Bayes classifier)**.
5. Discuss the **independence assumption** in Naïve Bayes

Question 1 — The Boy Who Cried Wolf: Identifying TP, FP, FN, TN

This case study is a classic illustration of a confusion matrix, where the shepherd boy's cry of "Wolf!" is treated as the model's PREDICTION, and the actual presence/absence of a wolf is the ACTUAL (ground truth) condition.

Mapping the story to a confusion matrix

Event	Prediction (boy's cry)	Actual condition	Confusion Matrix Category
Boy cries "Wolf!" (Day 1) but no wolf is present	Wolf present	Wolf NOT present	False Positive (FP)
Boy cries "Wolf!" (Night, real wolf appears) but villagers don't respond	Villagers predict: No wolf (they ignore him)	Wolf IS present	False Negative (FN)
(Hypothetical) Boy cries "Wolf!" and a wolf is actually present, villagers respond	Wolf present	Wolf present	True Positive (TP)
(Hypothetical) Boy stays silent and no wolf is present	No wolf	No wolf	True Negative (TN)

Interpretation

- FP (False Positive): The first alarm — the boy predicted a wolf when none existed. This is a "false alarm", i.e., Type I error.
- FN (False Negative): The second, real alarm — a wolf truly appeared, but because the villagers (the "classifier") had learned to distrust the signal, they predicted "no wolf" and failed to act. This is a missed detection, i.e., Type II error.
- TP and TN do not literally occur in the narrated story, but are included conceptually to complete the confusion matrix: TP would be a genuine wolf correctly reported and acted upon; TN would be a quiet night correctly left unreported.

Key takeaway: The story demonstrates the real-world cost of False Positives — repeated false alarms erode trust in the classifier (the boy), causing the villagers to later commit a costly False Negative when a true event occurs. In classifier design, this motivates the trade-off between Precision (avoiding FPs) and Recall (avoiding FNs).

Question 2 — Cat Image Classification: Confusion Matrix, Precision, Recall, F1

Given

- Total images = 100
- Actual cat images = 30, Actual non-cat images = 70
- Model correctly predicts "cat" in 25 of the 30 cat images
- Model correctly predicts "no cat" in 50 of the 70 non-cat images

Step 1: Build the confusion matrix

Treat "cat" as the Positive class and "no cat" as the Negative class.

- True Positive (TP) = correctly predicted cat = 25
- False Negative (FN) = actual cat, predicted no cat = 30 − 25 = 5
- True Negative (TN) = correctly predicted no cat = 50
- False Positive (FP) = actual no cat, predicted cat = 70 − 50 = 20

	Predicted: Cat	**Predicted: No Cat**	**Total**
Actual: Cat	TP = 25	FN = 5	30
Actual: No Cat	FP = 20	TN = 50	70
Total	45	55	100

Step 2: Precision, Recall, and F1 Score

Precision = TP / (TP + FP) = 25 / (25 + 20) = 25 / 45 = **0.5556 (≈ 55.56%)**

Recall (Sensitivity) = TP / (TP + FN) = 25 / (25 + 5) = 25 / 30 = **0.8333 (≈ 83.33%)**

F1 Score = 2 × (Precision × Recall) / (Precision + Recall)

= 2 × (0.5556 × 0.8333) / (0.5556 + 0.8333) = 2 × 0.4630 / 1.3889 = **0.6667 (≈ 66.67%)**

Interpretation

- High Recall (83.3%) means the model catches most actual cat images, missing only 5 of 30.
- Lower Precision (55.6%) means that among all images the model flags as "cat", a good portion (20 of 45) are actually false alarms (non-cat images misclassified as cat).
- The F1 Score (66.7%) balances both metrics and shows the model is moderately reliable but has room for improvement in reducing false positives.

Question 3 — Naïve Bayes Spam Classification

Given Dataset

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

1. Prior Probabilities

Total records = 6; Spam records = 3 (rows 1, 3, 5); Not Spam records = 3 (rows 2, 4, 6).

P(Spam) = 3/6 = 0.5

P(Not Spam) = 3/6 = 0.5

2. Conditional Probabilities for Each Feature

Raw (unsmoothed) conditional probabilities computed from the frequency table:

Feature = Value	P(Value | Spam)	P(Value | Not Spam)
Offer = Yes	2/3 = 0.667	1/3 = 0.333
Offer = No	1/3 = 0.333	2/3 = 0.667
Free = Yes	3/3 = 1.000	0/3 = 0.000
Free = No	0/3 = 0.000	3/3 = 1.000
Length = Short	2/3 = 0.667	1/3 = 0.333
Length = Long	1/3 = 0.333	2/3 = 0.667

Note the zero-frequency problem: P(Free = No | Spam) = 0, since every spam record in the sample has Free = Yes. A raw Naïve Bayes multiplication would force the entire Spam posterior to zero regardless of the other features, which is unrealistic. Laplace (add-one) smoothing is therefore applied below.

Laplace-Smoothed Conditional Probabilities

Using add-one smoothing with 2 possible values per feature: P(value|class) = (count + 1) / (class total + 2)

Feature = Value	**P(Value | Spam)**	**P(Value | Not Spam)**
Offer = Yes	(2+1)/5 = 0.60	(1+1)/5 = 0.40
Offer = No	(1+1)/5 = 0.40	(2+1)/5 = 0.60
Free = Yes	(3+1)/5 = 0.80	(0+1)/5 = 0.20
Free = No	(0+1)/5 = 0.20	(3+1)/5 = 0.80
Length = Short	(2+1)/5 = 0.60	(1+1)/5 = 0.40
Length = Long	(1+1)/5 = 0.40	(2+1)/5 = 0.60

3. Classifying: (Offer = Yes, Free = No, Length = Short)

Applying the Naïve Bayes assumption of conditional independence between features given the class:

P(Spam | X) ∝ P(Spam) × P(Offer=Yes|Spam) × P(Free=No|Spam) × P(Length=Short|Spam)

= 0.5 × 0.60 × 0.20 × 0.60 = **0.036**

P(Not Spam | X) ∝ P(Not Spam) × P(Offer=Yes|NotSpam) × P(Free=No|NotSpam) × P(Length=Short|NotSpam)

= 0.5 × 0.40 × 0.80 × 0.40 = **0.064**

Normalizing (dividing each by the sum 0.036 + 0.064 = 0.100):

- P(Spam | X) = 0.036 / 0.100 = 0.36 (36%)
- P(Not Spam | X) = 0.064 / 0.100 = 0.64 (64%)

Since **0.64 > 0.36**, the Naïve Bayes classifier predicts:

Predicted Class = NOT SPAM

4. Comparison with WEKA (Naïve Bayes classifier)

WEKA's NaiveBayes classifier (nominal attributes) applies Laplace estimation by default in the same way shown above, so running this exact dataset in WEKA (Explorer → Classify → NaiveBayes, 6 instances, nominal attributes Offer/Free/Length, class Spam/Not Spam) is expected to reproduce the same conditional probability table and the same predicted class, Not Spam, for the test instance (Offer=Yes, Free=No, Length=Short).

- Because the dataset is tiny (only 6 instances), WEKA's estimates will match the hand-computed Laplace-smoothed values almost exactly.
- Minor numerical differences (if any) can arise from WEKA's use of a small kernel/normal-distribution correction for numeric attributes — not relevant here since all three features are nominal — or from rounding in the displayed posterior probabilities.
- Students should confirm by loading the ARFF/CSV version of this dataset into WEKA and checking the "Predictions on test data" output for the instance in question.

5. Discussion: The Independence Assumption in Naïve Bayes

Naïve Bayes gets its name from the "naïve" assumption that all features are conditionally independent of one another given the class label. Formally:

$$P(Offer, Free, Length \mid Class) = P(Offer|Class) \times P(Free|Class) \times P(Length|Class)$$

- In reality, this assumption is often violated. For example, emails containing the word "Offer" are also very likely to contain the word "Free", since promotional/marketing spam frequently uses both terms together — i.e., these two features are correlated, not independent.
- Despite this violation, Naïve Bayes frequently performs well in practice (particularly for text/spam classification) because what matters for correct classification is usually only which class gets the higher posterior probability, not the exact probability values — the independence assumption tends to bias both class posteriors in a similar direction, so relative ranking is often preserved.
- The assumption also makes the model extremely fast to train and score, and robust on small or high-dimensional datasets (such as this 6-record dataset), which is why it remains popular for spam filtering even though the independence assumption is technically "naïve" or unrealistic.
- When features are strongly correlated, more expressive models (e.g., Logistic Regression, Decision Trees, or Bayesian Networks that explicitly model dependencies) may yield better-calibrated probabilities, though not always better classification accuracy.